

Bathroom Trash Can Project
February, 27th 2024
Ms. Ross

Presentation and Investigation Part A

1) Who is involved in this problem?

A wide range of people are the intended audience for this design. Although it is evident that all women who use the restrooms in GPS—including teachers, students, and any female visitors—are a major target, it also has an impact on the women who clean our restrooms and remove product waste.

2) What are they doing?

There are women using the restrooms. School time is fast-paced, so all women in the school have a short amount of time to properly use the restroom. With that, women are faced with handling whatever personal belongings they are carrying at the same time, judging by a lack of pockets in our traditional uniforms and women's clothing in general. Women's phones, keys, and other portable goods must be kept on a shelf in the bathroom while they use the restroom, which is quite inconvenient, or they must be placed in an unstable and unsanitary spot. If on her menstrual cycle, the woman is forced to discard it in our old metal containers which drop to a shared box between multiple stalls. The woman will have to leave the cubicle during her menstrual period and obtain supplies before going back to the stall to treat her cycle if she discovers she is on it while she is physically in the stall.

What unique situations does the problem occur within, if any?

If the woman discovers she is on her menstrual cycle while she is physically in the stall, she will have to leave her stall during her menstrual period and obtain supplies before going back to the stall to treat her cycle. This is not only unsanitary but is uncomfortable and runs the risk of getting blood on her clothing. Additionally, placing phones, keys, or any other personal items on an unsanitary place in the stall will contaminate it, and with the fast paced school day girls don't have time to properly clean them from these germs obtained in the bathroom stall.

3) What is the size of the issue?

This issue is particularly noticeable to me since I attend an all-girls school, but it is also present in other classrooms and public facilities. Not only have the three of us experienced it personally, the female community's social media presence has made it clear as well.

4) Where is the problem happening?

Our design is centered on the bathroom issue that is present in the Girls Preparatory School community. Since our feminine products are situated near the bathroom entrances, the issue affects every bathroom. Another problem with needing to carry loose personal belongings while using the restroom is that there isn't a ledge or a clean place to put them, so people frequently end up placing them on the shaky toilet paper holders or metal bars in the handicap restrooms. Finally, we wanted to address the problem of how to dispose of our menstrual products because they are not only disgusting to touch but also violate privacy because of the connecting holes that allow you to see into neighboring

stalls.

When does the problem occur?

When a female at Girls Preparatory School is using the restroom. Both in the event of a normal bathroom break or during the time of menstruation.

5) Why does the problem happen?

As stated above, the plainness in the stalls is the cause of the issues. Because there is nowhere to store feminine products inside the stalls, products are placed outside out of reach, typically at the entrance of the bathroom. Additionally, women are usually seen carrying loose items around school, and our dresses, uniforms, and overall women's clothing lack prepared pockets for holding items while using the restroom—especially since you have to slide them down. Overall, these factors contribute to the placement of products outside of each restroom.

How often does the problem happen?

This is a constantly recurring issue. From hour to hour.

Stakeholder Concerns and Connections (A3, A4)

Directly indicate at least five primary stakeholders or stakeholder groups. These are the people most directly impacted by the problem. Highlight the concerns of your primary stakeholders and use that to justify why there is a need to address the problem. Consider the breadth of the problem and how it might have social and environmental impacts on secondary stakeholders, which are people who may be impacted by a solution to the problem and are still part of the system.

1. Student Body at GPS
2. Faculty /Teachers at GPS
3. Maintenance/Cleaning persona at GPS

This includes the necessary stakeholders for this project. These are the primary constant users of the bathrooms here at GPS.

Problem Importance (A6)

End Element A with a compelling argument for why this problem is important and needs to be solved now. Support this argument with data and statistics.

We believe that a problem that began as a search for a suitable ledge or space for women to store belongings while using the restroom for w at Girls Preparatory School could be expanded upon in order to address additional issues, such as the privacy concerns associated with some current systems of disposing of feminine products in the restroom stalls and the feminine products that are not accessible while in the stall. We wanted to be mindful of this even though it wasn't the primary concern while we were creating our project and chose to utilize more antibacterial materials than the metal on the existing trash cans in the restrooms. Furthermore, we

wanted to avoid putting our cleaning crew under any more strain when they were cleaning our project and replacing the feminine product containers.

In our community our stakeholders (female teachers and students at Girls Preparatory School) showed in a survey that 85% of students surveyed claimed that they wish they had a cleaner space in a stall to place their items while using the restroom. With that, 89% of students surveyed claimed that they have been placed in a situation in which they needed a feminine product while in the stall, but could not access it or have someone to access it for them at the moment. Lastly, 70% of students surveyed claimed that they believed that there could and should be a cleaner solution than the swinging doors that we have now. Not to add the well-known deficiency of pockets in women's apparel. According to The Badger Gerlarald's article on "The History behind the absence of pockets from traditionally female clothing," today's women's pants have pockets that are, on average, 6.5% smaller and 48% shorter than those of men's pants. This is an issue at GPS since most instructors and kids carry a phone, keys, or other small portable items when using the toilet rather than a handbag while at school.

Overall, in an attempt to make a ledge/space to store personal items, we decided to add other aspects to contribute to other issues such as a lack of accessibility with feminine products while in the stalls, making the feminine trash easier to change for the cleaners while also using materials to make it more antibacterial as well.

Documentation and Analysis of prior attempts Part B

Prior Design Attempt #1 - <original metal can>

Source

<https://www.webstaurantstore.com/documents/specsheets/179b254.pdf>



Details

- Metal trash can
- Push in swinging door
- Connected through stall wall
- Shared trash can for both stalls

Analysis

The metal surface harbors bacteria, and the swinging door requires contact for the trash can to be used. This is extremely unsanitary, especially considering the contents of which are being disposed of. Additionally, the trash can is shared between stalls. This becomes a privacy issue as the user in stall 1 can easily view the user in stall 2 through the opening.

Prior Design Attempt #2 - <plastic can>

https://www.webstaurantstore.com/lavex-janitorial-white-plastic-wall-mount-sanitary-napkin-recptacle-imp-1102/2721102.html?utm_source=google&utm_medium=cpc&utm_campaign=GoogleShopping&gclid=CjwKCAiAu9yqBhBmEiwAHTx5p4VL5_4ifyyOTbR-_5Vklpf9HkbWyCevX62lbKHx8HEe6yjqEaMNDxoC-8QQAvD_BwE

Journal/newspaper about feminine product trashcans and dispensers page 2

https://nsuworks.nova.edu/cgi/viewcontent.cgi?article=1556&context=nsudigital_newspaper/

<https://www.5gyres.org/plastic-period-care>

Info about use of plastic period product

https://www.googleadservices.com/pagead/aclk?sa=L&ai=DChcSEwjL1ob0zMuCAxWBNtQBHUkrB70YABARGgJvYQ&ase=2&gclid=CjwKCAiAu9yqBhBmEiwAHTx5p2QiFVV1y8xpkhjuk5bN70SS8q3qL8Q5MlvsV-VB9EfpdpxVOYXMmhoCERsQAvD_BwE&ohost=www.google.com&cid=CAESVuD2-qn6O7zUfggP63Y8haWa32UpV1ewmmpFDXQWP6h9pBdudeQTe6Dr5yQQZtowJatqaJEbF8KNdvjn2ti2QuwxYaAZAoqau8TdrOn6-oFWgk924gLR&sig=AOD64_0ku2PvOyLg98fOgiZ_YO21LKJjMA&q&nis=4&adurl&ved=2ahUKEwjdyfnzzMuCAxXXrmoFHatwAk4Q0Qx6BAgMEAE



Include main details about this prior design attempt

The design attaches to a stall in a bathroom in a public setting. There is a simple lid that opens to a wax trash bag where they can put their products. This simply screws into the wall and is made out of plastic for easy cleaning.

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

What is successful about this design is that it is easy for staff members of the school to change out the bag and clean it. What we think could be improved is that the cleanliness is not there. Every person must touch the lid to allow the disposal of their products. We believe there should be a different way to do it. Although this is accessible, there is room for improvement. *The door is also not spring-loaded which means the person will have to touch the thing many times*

Presentation justification and solution of design requirements Part C

Design Requirement #1 (HIGH Priority)

Our design should be antibacterial to allow for easy and clean access to feminine product trash cans without contamination. R

Explain how this design requirement directly ties to your prior research.

Our prior research indicates that there are many different types of bacteria found on bathroom surfaces- some harmless, and others very concerning, including but not limited to: E. Coli, Salmonella, and Staphylococcus Aureus.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

Design Requirement #2 (HIGH Priority)

Our design should provide a clean and practical space for feminine products to be stored within a bathroom stall

Explain how this design requirement directly ties to your prior research.

While observing several restrooms around campus, we realized that none had easy access to feminine products from within the stalls. All had feminine products stored outside the stalls near the sinks but our design intends to enhance the bathroom experience by storing them within the individual toilet stalls.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

Design Requirement #3 (HIGH Priority)

Our design should provide a clean ledge/shelf to store any handheld items within a stall when using the bathroom .

Explain how this design requirement directly ties to your prior research.

After observing the bathrooms around campus and realizing personal complaints about a lack of pockets and/or places to put phones, keys, and other items when using the bathroom, we realized that this universal issue can be easily resolved with a small shelf or space within the stall.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

Design Requirement #4 (HIGH Priority)

Our design should be spatially efficient, with enough room to hold intended items while still being conservatory and non-bulky.

Explain how this design requirement directly ties to your prior research.

No research- stalls are just a compact area and we have to be spatially aware.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

Design Concept Generation, Analysis, and Selection Part D

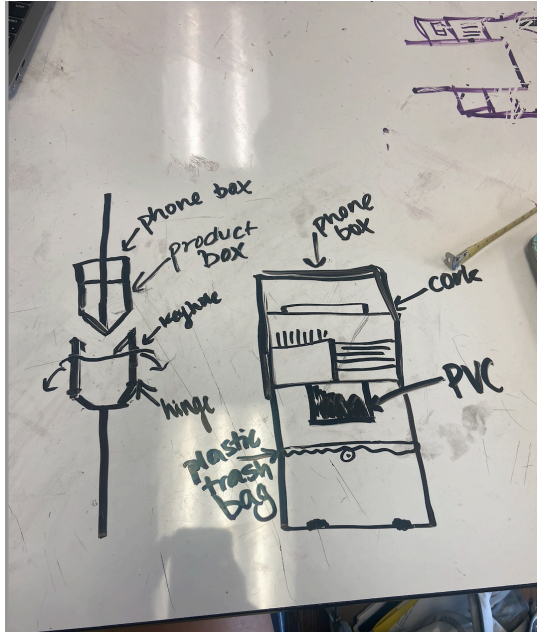
Design Idea Generation (DI)

This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as needed.

Our first step was drawing out our ideas for possible prototypes. We then researched materials as well as what is currently on the market. After this, we put together a very rough cardboard prototype and took it to Mr. Glass for official prototype planning. Over the next few weeks we worked closely with him to laser cut and assemble pieces of plexiglass to create the final prototype.

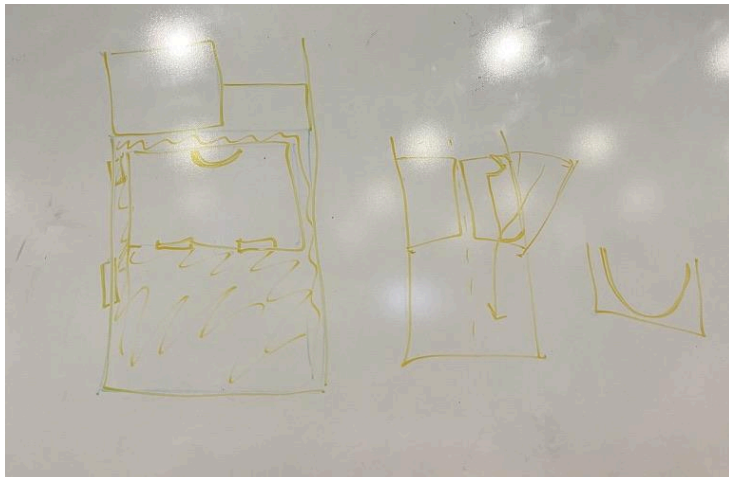
Design Idea #1

Date:



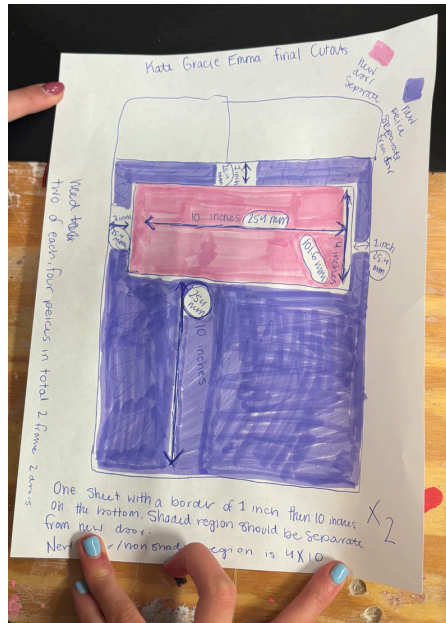
Design Idea #2

Date: 2/28/24



Design Idea #3

Date: 3/4/24



Design Blueprint (D4)

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Design Blueprint #1

Manufacturing Methods:

Step 1: Laser cut pieces of plexiglass & paint them

Step 2: Use adhesive/tape to assemble the pieces

Step 3: Apply cork to the bottom of the shelving

Item	Cost each	# needed	Total
Plexiglass	Free (recycled)	2 sheets	\$0
Cork sheets	About \$0.75	4 sheets	\$3
PVC Pipe	\$9	0.5 of a whole pipe	\$4.50
Total Cost			>\$9

Application of stem principles and practices Part E

STEM Principle for High-Priority Design Requirement #1

STEM Concept: [Micro-biology](#)

Describe how STEM concept applies directly to the specifics of your design:

[As our design is intended for bathrooms, it is exposed to microbiomes and prone to collection of bacteria](#)

Proposed Solution to STEM Principles and Practices (E2)

STEM Principle for Design Solution #1

STEM Concept: [Microbiology](#)

Describe how STEM concept applies directly to the specifics of your solution:

[We utilized materials \(cork and pvc\) that demonstrate antibacterial properties.](#)

Consideration of design viability part F

Design Viability Description	Connection to Design Requirement	Credible Evidence?
The solution is inexpensive	Design does not cost more than \$100	After ordering all parts our cost of production was less than \$50
The solution is anti-bacterial	Uses cork and PVC	Testing in progress
The solution is practical	Provides a space to place items and has easy in-stall access to feminine products	Survey results (majority- 70%+ students surveyed felt a new solution should be developed)

Construction of a testable prototype part G

Ensure that photos are taken from multiple angles so that all sides and subsystems can be seen. All images should have captions and should be referenced in-text.

Prototype Construction of Design #1 (make as many copies, updating the design number, as needed)

Date:

Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:

Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

	Details	Tools and Manufacturing Methods used
Step 1	Gather materials	Spare cardboard from boxes, duck tape, and scissors
Step 2	Cut pieces of cardboard	Cardboard and scissors/exacto knife
Step 3	Tape pieces together to form the 3 basic pieces of the design (trash can, product shelf, personal item shelf)	Tape and hands!

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

Final Prototype Construction of Design #2 (change this number to match the design number you're on)

Date:

Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:

Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

	Details	Tools and Manufacturing Methods used
Step 1	Cutting plexiglass	Laser cutter
Step 2	Apply superglue and secure pieces	Super glue and catalyst
Step 3	Attach doors to sides with ducttape	duct tape
Step 4	Glue pvc	PVC attached with plexiglass glue and used sandpaper.
Step 5	PVC was cut before previous step	With power saw
Step 6		
Step 7		

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

List of material details, including item name, description, where it was purchased/obtained (include URL if applicable), cost per unit (if applicable), # of units, total cost (if applicable)

Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item
PVC pipe	Pipe used as a slide for products	https://www.amazon.com/Schedule-PVC-Pipe-Custom-Lengths/dp/B0C5DW4ZP4/ref=sr_1_1_sspa?crid=3VRO873W8KGV&keywords=3%2Bpvc%2Bpipe&qid=1704728749&sref=3%22%2Bpvc%2Caps%2C148&sr=8-1-spons&p_csd=d2kZ2V0TmF9ZT1zcF9hdGV&th=1	8.97 per unit	2	17.94
Cork strips	Used for cleanliness	https://www.amazon.com/dp/B0B66S1WVY?ref=cm_sw_r_cso_sms_apin_dp_X2TW0R8NJBKVR5V56PX8&cafhIReslm_g=1&language=en-US&th=1	7.99	2 packs	15.98
Plexiglass adhesive glue	To secure everything together	https://www.amazon.com/Acrylic-Adhesive-Materials-Materials-Products-Transparent/dp/B0B972J186/ref=ase_df_B0B972J186/?tag=hvprod-20&linkCode=df0&hvadid=642056799018&hvpone=&hvptwo=&hvqmt=&hvdcmid=&hvllocint=&hvllophy=9013299&hvtargid=pla-1949705321753&psc=1&mcid=b1d816ca274f3948941f41015ab8fd6	4.16	6	25.00

Testing or Modeling of Attributes (G3, G4) and Addressing Non-Testable Attributes (G4)

Describe all of your prototype's attributes (features, aspects, details, or subsystems). Demonstrate that your prototype is constructed with sufficient functionality that your prototype can test all attributes of your design. This requires you to look back at your initial proposed design details from Element D.

For example, if your initial design details called for wheels and motors, is your prototype sufficiently functional such that you can test out those wheels and motors?

If there are aspects of your design that you believe cannot be physically prototyped or mathematically modeled within the scope and limits of your class, you must later provide evidence that an expert has reviewed your justifications and that you have incorporated their feedback. You do not need to conduct that review in Element G. For now, enter “expert review required” for these attributes in the final column.

Write in this after the experiment is done to verify

Prototype testing of data collection plan Part H

	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1	Our design should be antibacterial to allow for easy and clean access to feminine product trash cans without contamination.	We formulated an experiment using agar trays. We poured 6 total agar trays to test. We concluded that of the 6 trays, two trays will be a control variable to receive baseline data, 2 will be a before and after in an upper school bathroom, and the last two will be a before and an after in a middle school bathroom. (We chose the busiest bathrooms for each building).	Test case: 1 Trials: 2	N/A	Fail: MORE bacteria is evident on our model Pass: Less or equal bacteria is on our model Goal: Significantly less bacteria evident on our model
2	Our design should provide a clean and practical space for feminine products to be stored within a bathroom stall	Using all of the feminine products provided by GPS, we will try to fit them into our shelf and ensure that it is not too big or too small.	Test case: 1 Trials: 1	N/A	Fail: Doesn't fit into the shelf or falls out of the shelf Pass: Fits comfortably in the shelf (possibly a little bit of a squeeze or a little too much room) Goal: Perfect fit, products are also easy to put in and take out
3	Our design should provide a clean ledge/shelf to store any handheld items within a stall when using the bathroom.	Place a phone and keys on the shelf and observe whether they fit comfortably, tight, or roomy.	Test case: 1 Trials: As many different-sized phones as possible	N/A	Fail: 1 or more phones do not fit on the shelf Pass: All phones fit on the shelf (possibly a little tight but still fitting) Goal: All sizes of phones fit comfortably and are easily placed in and taken out

					off the shelf
4	Our design should be spatially efficient, with enough room to hold intended items while still being conservatory and non-bulky.	Take a random sample of ** Individuals, ensuring that a range of heights, weights, and overall sizes are represented in the sample group. After having them walk in, and use the restroom as usual with our prototype in the stall, send them a survey to express all areas of concern with the prototype.	Test Case: 1 Trials: 1 trial with each individual selected	N/A	Fail: Users find it is too bulky Pass: No complaints from users Goal: hardly notice it is there!

Examples of testing Part I

Tests

1)

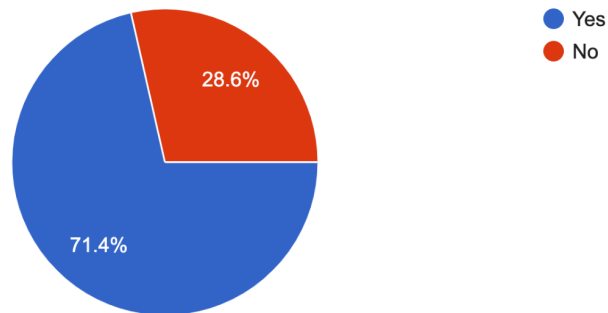
Swab tests from different feminine product disposals

Swab number	Amount of bacteria
Og bathroom design	In progress
Our bathroom design	In progress

2) Survey of senior class for experience in a bathroom

Have you ever felt like the stainless steel swinging door disposal trashcans either could be more clean or that there should be a more accessible cleaner option?

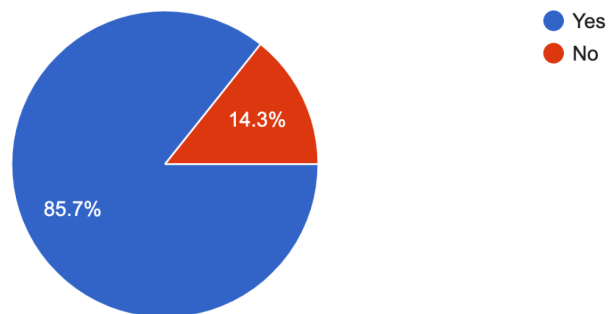
28 responses



74 percent of the senior class say they feel that the stainless steel trashcans could be more clean or there should be a cleaner more accessible option.

Have you ever been in a bathroom stall and felt that you wished there was a clean place to put your phone or backpack while you use the restroom?

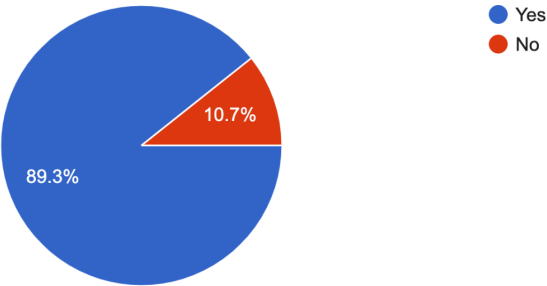
28 responses



85.7 percent of the senior class feels like they wished there was a clean place to put their belongings in the stall.

Have you ever felt that you needed a tampon or pad while in the stall and are unable to leave the stall and get one?

28 responses



89.3 percent of the senior class says they have been in the bathroom stall and were unaware they would need a feminine product and then could not leave the stall to go get one.

Some quotes from the senior class expressing distaste for the bathroom stall disposal trashcans include, “the dinky lil plastic trash holders in the bathroom for tampon/pad disposal. IDK how we could efficiently and cheaply fix that, but it’s the nastiest thing to me”

“Bathrooms are freaking gross I don’t like the tampon trash can ”

3) Phone sizes

iPhone sizes	inches
iPhone 15 Pro max	6.69 X 3.08
iPhone 15 pro	5.77 X 2.78
iPhone 15 Plus	6.33 X 3.06
iPhone 15	5.81 x 2.82
Samsung galaxy Z fold	7.6 unfolded